UNIT 07 · TOPIC 04 · T-TESTS FOR MEANS

t-Tests for Means & Paired Data

Why we use t when σ is unknown, one-sample, matched-pairs, and two-sample t-test formulas, degrees of freedom, and conditions.

1. Why t Instead of z? & Variable Definitions

In real problems, the population standard deviation σ is almost always **unknown**, so we replace σ with the sample standard deviation s. Because s adds extra variability from sample to sample, the standardized statistic follows a **Student's t-distribution** with heavier tails than $N(0, 1)$. As $df \to \infty$, $t(df) \to N(0, 1)$.

SYMBOL	NAME	DEFINITION & NOTES
$\bar{x}, \mu_0$	Sample Mean & Null Mean	$\bar{x}$ = observed sample mean; μ_0 = hypothesized population mean under H_0.
s	Sample Standard Deviation	$s = \sqrt{\frac{\Sigma(x_i - \bar{x})^2}{n-1}}$. Plugged in for unknown σ.
df	Degrees of Freedom	Controls tail thickness of t-curve. $df = n - 1$ for 1-sample / paired; Welch's df for 2-sample.
$d_i, \bar{d}, s_d$	Paired Differences	$d_i = x_{1i} - x_{2i}$ for pair i; $\bar{d}$ = mean of differences; s_d = SD of differences; n = number of pairs.
μ_d	True Mean Difference	Population mean of paired differences ($\mu_d = \mu_1 - \mu_2$). Null is usually H_0: $\mu_d = 0$.

2. The Three t-Test Formulas

1-SAMPLE T-TEST FOR A POPULATION MEAN (H_0: $M = M_0$)

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}} \quad \text{with} \quad df = n - 1$$

Normal / Large Condition: Population is Normal, OR $n \geq 30$ (CLT), OR sample data plot shows no strong skewness or outliers (if $n < 30$).

MATCHED PAIRS T-TEST (H_0: $M_D = 0$)

$$t = \frac{\bar{d} - \mu_{d,0}}{s_d / \sqrt{n}} \quad \text{with} \quad df = n - 1 \ (n = \text{number of pairs})$$

Key Idea: Compute the difference $d_i = \text{After} - \text{Before}$ for every pair first, then run a standard **1-sample t-test** on the single list of differences!

2-SAMPLE T-TEST FOR INDEPENDENT MEANS (H_0: $M_1 - M_2 = 0$)

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)_0}{\sqrt{\frac{{s_1}^2}{n_1} + \frac{{s_2}^2}{n_2}}}$$

Degrees of Freedom (Unpooled / Welch's): Use technology's exact Welch df, or conservative $df = \min(n_1 - 1, n_2 - 1)$ by hand. **Never pool** sample variances for 2-sample t unless explicitly told $\sigma_1 = \sigma_2$.

3. Choosing Correctly: Paired t vs. 2-Sample t

QUESTION TO ASK	MATCHED PAIRS T-TEST	2-SAMPLE T-TEST
Are the two measurements linked?	**Yes** — Same subject measured twice (Before/After), twins, or matched pairs.	**No** — Two completely separate, independent random samples or treatment groups.
Sample Sizes	Must have equal counts ($n_1 = n_2 = n$ pairs).	n_1 and n_2 can be different.
Degrees of Freedom	$df = n - 1$	Welch df (up to $n_1 + n_2 - 2$) or $\min(n_1 - 1, n_2 - 1)$
Normal Condition Checked On	The single list of differences d_i.	**Both** Sample 1 and Sample 2 separately!

4. Calculator Syntax & Common Traps

COMMON MISTAKES

- Using a **2-sample t-test** on paired Before/After data (violates independence between the two groups!).
- Checking $n_1 + n_2 \geq 30$ instead of checking **each** sample $n_1 \geq 30$ AND $n_2 \geq 30$ separately.
- Using a z-test when given sample SD s instead of population SD σ.

TI-84 / NSPIRE COMMANDS

- **p-value from t:** `tcdf(lower, upper, df)`
- **Critical value t^*:** `invT(area_left, df)` (e.g., 95% CI → `invT(0.975, df)`)
- **Menus:** `T-Test` (1-sample & paired on diff list), `2-SampTTest` (unpooled: `Pooled: No`).